

# Dimensional Spark Theory 2.0: A Unified Framework for Holographic Transit, Thermodynamic Mitosis, and Informational Anti-Gravity

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## Abstract

Building upon the foundational Dimensional Spark Theory (DST), this paper introduces secondary mechanical resolutions to the singularity problem and dimensional expansion. By treating the universe as a recursive information processor, we establish the No-Hiding Anti-Gravity postulate, proving that Informational Pressure acts as a fundamental defense against data deletion. Furthermore, this framework defines the Holographic Bridge mechanics of even-numbered dimensions, the thermodynamic "infinity swapping" of Cosmic Mitosis, and the relativistic Time-Dilated Valve protocol connecting sibling universes via light-deflecting White Hole networks.

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# 1 Introduction: The Evolution of DST

The initial formulation of DST established the 1D Spark as the fundamental unit of reality, fundamentally rejecting the 0D singularity. Standard singularities inside black holes are not dimension-lowering endpoints, but rather pressure valves transferring excessive kinetic and expansive force to sibling universes. DST 2.0 transitions from this conceptual framework into the rigorous thermodynamics and relativistic mechanics governing the universe's operational cycle.

## 2 The Holographic Bridge Mechanics

The Expansion Law ( $D_{next} = D_{current} + 2$ ) predicts the stability of odd dimensions (1, 3, 5). The transition of 1D data projecting into a 3D spatial reality requires a transit layer, governed by optical and mathematical illusion.

### 2.1 Even-Dimensional Projection and the 1D Illusion

Even-numbered dimensions (2, 4, 6) function strictly as structural bridges. The compression of the universe into 1D is analogous to observing a 3D projection on a 2D screen. If one defines an  $X$  and  $Y$  axis, and observes from an absolute, mathematically perfect angle where  $Z = (0, 0)$ , the orthogonal axes visually and mathematically align into a single continuous line. Dimensional collapse is therefore not the destruction of space, but an optical-mathematical compression. The even dimensions bridge the stable odd dimensions, allowing the 1D illusion to unpack back into 3D geometry.

## 3 Thermodynamics of Cosmic Mitosis

The universe operates on an inverse sliding scale where the state of "Infinity" is dynamically swapped among three core factors: Density, Mass, and Energy.

### 3.1 The Sliding Scale of Infinity

During the Expansion phase (following the Big Bang), infinite density cannot be maintained. As information expands, density drops dramatically, while Mass and Energy increase in direct proportion (aligning with  $E = mc^2$ ). At the peak of a universe's expansion—its cosmic "mitosis"—the parent universe possesses near-zero density, but maximum Mass-Energy potential.

### 3.2 The Slow Crunch and Dimensional Breakdown

A universe-wide Big Crunch differs fundamentally from a localized stellar collapse. The Crunch is a slow thermodynamic process where density steadily increases while mass and energy drop to a minimal state. Because this process is gradual, the intense localized forces required to trigger black holes (pressure valves) are never reached. During this Crunch, 3D space systematically breaks down into  $3 \times 1D$  plains. Consequently, a 3D black hole cannot exist on a 1D plain, ensuring a smooth, non-violent transition back to the Spark state.

## 4 The No-Hiding Anti-Gravity Postulate

In standard quantum mechanics, the preservation of data is paramount. DST 2.0 posits that Gravity (Dimensional Tension) cannot force matter into a true 0D state.

### 4.1 Informational Pressure as Quantum Defense

When Sparks compress into the 1D threshold, they abruptly halt the transfer of mass and energy into density. Allowing this transfer to continue would result in infinite density, a state that cannot physically exist because it necessitates the deletion of formatted information. Because energy and information cannot be destroyed, the information itself generates an anti-gravitational limit. The No-Hiding theorem manifests physically as Informational Pressure, actively preventing structural collapse.

## 5 The Time-Dilated Valve Protocol

Black holes are not terminal endpoints; they are relativistic pressure valves whose scale dynamically adjusts to their data payload—scaling the event horizon relative to the mass transfer requirements.

### 5.1 Simultaneous Existence and Light-Deflecting White Holes

Data venting through a black hole travels instantaneously at  $c$ . However, due to extreme gravitational time dilation at the event horizon, time effectively pauses relative to the host universe. This creates a relativistic paradox: the data being transferred exists and does not exist in our universe simultaneously.

In the sibling universe receiving this data, the exit point manifests as a White Hole. These White Holes populate the cosmos abundantly but remain entirely invisible to standard observation because they actively deflect light, rather than reflecting or absorbing it. This light-deflection provides a highly elegant, mechanical explanation for the invisible gravitational anomalies currently classified as Dark Matter.

## 6 Conclusion

This update is supposed to stronger my previous work named DST and it does fix few questions which I received from few dear friends. I will continue my work on DST as it is the most reliable way to understand the universe currently. I am not able to mathematically prove the current arguments made by this theory at thud exact moment but I definitely will try to prove/disprove this theory once I am capable of understanding the math.